Statistical Issues in Multicenter Randomized Clinical Trials

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April 19, 2026

Contents

1 Introduction

Multicenter randomized clinical trials (RCTs) represent the standard design for confirmatory evaluation of therapeutic interventions. By enrolling participants across multiple clinical sites, these trials achieve practical recruitment targets while broadening the population base for generalization of findings[1,2]. However, the multicenter structure introduces several analytic complexities that, if mishandled, can distort inference about treatment effects. This report addresses four interrelated statistical issues that arise in the design and analysis of multicenter RCTs with continuous outcomes.

1.1 Modeling site effects

The first question concerns how site (or center) effects should be incorporated into the primary analysis. Three broad strategies exist: ignoring site differences entirely, treating sites as fixed effects, and treating sites as random effects drawn from a population of potential sites. Each strategy carries distinct assumptions and implications for the validity and efficiency of treatment effect estimation[3,4].

[2] provided an influential overview of the problem, identifying three analytic challenges inherent to multicenter data: within-center correlation of outcomes, confounding by center, and effect modification across centers. The choice between fixed and random effects models has been debated extensively. Fixed effects models condition on the observed sites and make no distributional assumptions about site effects, but they consume degrees of freedom rapidly when the number of sites is large relative to total sample size[5]. Random effects models treat site effects as realizations from a normal distribution, estimating only a variance component rather than individual site parameters, and thereby preserve statistical efficiency[6]. The ICH E9 guideline recommends that mixed models are 'especially relevant when the number of sites is large'[1].

[7] conducted a comprehensive simulation comparing six analytic approaches for continuous outcomes across varying numbers of centers, center sizes, and intraclass correlation coefficients (ICCs). All methods yielded unbiased point estimates, but ignoring centers inflated the standard error and reduced power when within-center correlation was present. The mixed-effects model attained nominal coverage and near-optimal power across nearly all configurations. Generalized estimating equations (GEE) underestimated the standard error when the number of centers was small.

[8] extended these findings, demonstrating that random center effects performed at least as well as fixed effects in all scenarios examined, and substantially outperformed them when the number of patients per center was small.[9] confirmed these patterns in a recent replication study with both continuous and binary outcomes, reinforcing the recommendation for random effects models as a default analytic strategy.

1.2 Failure to adjust for stratification variables

The second issue concerns the consequences of ignoring stratification variables, including site, in the analysis. When randomization is stratified by center (as is standard in multicenter trials), failing to adjust for the stratification factor in the analysis leads to biased inference.[10] demonstrated through simulation that an unadjusted analysis of a trial randomized with stratified blocks produces standard errors that are biased upward, confidence intervals that are too wide, type I error rates below the nominal level, and a loss of statistical power. This paradoxical conservatism arises because the stratified randomization induces negative correlation between treatment groups within strata; ignoring this correlation inflates variance estimates.

[11] extended this work to trials with multiple stratification factors, finding that adjusting for the main effects of all stratification variables was generally sufficient without requiring adjustment for all cross-classified strata. These findings align with the ICH E9 recommendation that 'the statistical model used for analysis should reflect the restriction imposed by the randomisation procedure'[1].

1.3 Treatment-by-site interaction

The third concern is heterogeneity of the treatment effect across sites. Even under a common protocol, eligible participants, site practices, and investigator experience may vary, producing genuine treatment-by-site interaction[12].[3] argued that some degree of treatment-by-center interaction is inevitable in practice and that a rational approach to planning multicenter trials should anticipate it.

The ICH E9 guideline recommends first fitting a model without the interaction term to estimate the main treatment effect, and then examining heterogeneity as a secondary analysis. However, the power to detect treatment-by-site interaction is typically low, so a non-significant interaction test does not establish homogeneity[13].[4] compared fixed-effects weighted estimators, unweighted estimators, and random effects estimators in the presence of varying degrees of treatment-by-center interaction. The fixed effects weighted estimator performed well across a range of interaction magnitudes, while unweighted estimators could be inefficient when site sizes differed markedly.

When treatment-by-site interaction is modeled as random (i.e., the treatment slope varies across sites), the mixed model estimates both the average treatment effect and the variance of treatment effects across sites, providing a natural summary of heterogeneity[14].

1.4 Imbalanced site sizes

The fourth issue, closely related to the preceding three, is whether imbalance in site sizes affects inference. Multicenter trials frequently exhibit highly unequal enrollment across sites, with a few large sites contributing a disproportionate share of participants.[3] noted that a rational approach to planning multicenter trials leads naturally to unequal site sizes, because sites with higher recruitment capacity will enroll more patients within the trial's fixed time window.

[15] introduced a coefficient of imbalance to quantify the power loss attributable to unequal center sizes and showed that substantial imbalance can reduce power when an unweighted (Type III) analysis is used.[16] formalized this through a design effect framework, demonstrating that the efficiency loss depends on both the ICC and a statistic quantifying the heterogeneity of group distributions across centers. When the ICC is positive, balanced allocation is optimal, but moderate imbalance imposes only a modest efficiency penalty. The key risk emerges when site size is informative – that is, when larger sites systematically differ from smaller sites in patient characteristics or treatment delivery – potentially introducing bias into the treatment effect estimate[12,17].

1.5 Present study

The existing literature provides clear theoretical guidance but limited integrated simulation evidence spanning all four issues simultaneously. This report presents a Monte Carlo simulation that jointly varies the analytic method (ignore site, fixed site effects, random site effects), the presence or absence of treatment-by-site interaction, balanced versus imbalanced site sizes, and the magnitude of site-level variance. The simulation evaluates bias, empirical standard error, model-based standard error, power, and coverage probability of 95% confidence intervals for the average treatment effect across all factorial combinations of these design features.

2 Methods

2.1 ADEMP structure

Following Morris, White, and Crowther[18], the simulation is reported under the ADEMP framework.

- **Aims.** Compare three analytic strategies for handling site heterogeneity in multicenter RCTs (ignore site, fixed site effects, random site effects) across scenarios varying the number of sites, site variance, treatment-by-site interaction, and site-size balance.
- **Data-generating mechanism.** Detailed in the next subsection.
- **Estimand.** The overall treatment effect δ on the outcome scale.
- **Methods.** OLS ignoring site, OLS with fixed site effects, and a linear mixed model with random site effects via `lme4::lmer`.
- **Performance measures.** Bias, empirical SE, MSE, mean model SE, power at $\alpha = 0.05$, coverage of 95% CIs, and convergence rate, each reported with its Monte Carlo SE per Morris Table 6. Non-convergence is treated as a first-class outcome and its rate is reported per scenario and method.

2.2 Data generating model

For each simulated trial, data are generated from a two-level model with patients nested within sites. Let Y_{ij} denote the continuous outcome for patient j at site i:

$$Y_{ij} = \alpha_i + (\delta + \gamma_i)T_{ij} + \epsilon_{ij}$$

where $\alpha_i \sim N(0, \sigma^2_\alpha)$ is the random site intercept, $\delta = 0.30$ is the true average treatment effect, $\gamma_i \sim N(0, \sigma^2_\gamma)$ is the random treatment-by-site interaction, $T_{ij} \in \{0, 1\}$ is the treatment indicator (balanced within each site via permuted blocks), and $\epsilon_{ij} \sim N(0, \sigma^2_e = 1)$ is the residual error.

2.3 Parameter values

The simulation crosses the following factors:

- **Number of sites:** $K = 10$ (few) vs $K = 30$ (many)
- **Site-level variance:** $\sigma^2_\alpha \in \{0, 0.10, 0.50\}$ (none, moderate, large)
- **Treatment-by-site interaction:** $\sigma^2_\gamma \in \{0, 0.04\}$ (none vs present)

- **Site size balance:** balanced ($n_i = 20$ for all i) vs imbalanced (site sizes drawn from a Gamma$(0.8, 0.8/20)$ distribution, yielding a coefficient of variation ≈ 1.1)

The true treatment effect is $\delta = 0.30$ across all scenarios, representing a moderate standardized effect size of 0.30 standard deviations.

2.4 Analytic methods

Each simulated dataset is analyzed by three methods:

1. **Ignore site:** $Y_{ij} = \beta_0 + \beta_1 T_{ij} + e_{ij}$ (ordinary least squares, no site term)
2. **Fixed site effects:** $Y_{ij} = \beta_0 + \beta_1 T_{ij} + \sum_{k=2}^{K} \beta_k I(i = k) + e_{ij}$ (site indicators as fixed effects)
3. **Random site effects:** $Y_{ij} = \beta_0 + \beta_1 T_{ij} + u_i + e_{ij}$ where $u_i \sim N(0, \sigma_u^2)$ (linear mixed model via REML, fit with `lme4`)

2.5 Simulation procedure

Scenario 1/16: K10_noVar_bal ... Scenario 2/16: K10_modVar_bal ... Scenario 3/16: K10_hiVar_bal ... Scenario 4/16: K10_modVar_imbal ... Scenario 5/16: K10_hiVar_imbal ... Scenario 6/16: K10_modVar_int ... Scenario 7/16: K10_hiVar_int ... Scenario 8/16: K10_hiVar_int_imb ... Scenario 9/16: K30_noVar_bal ... Scenario 10/16: K30_modVar_bal ... Scenario 11/16: K30_hiVar_bal ... Scenario 12/16: K30_modVar_imbal ... Scenario 13/16: K30_hiVar_imbal ... Scenario 14/16: K30_modVar_int ... Scenario 15/16: K30_hiVar_int ... Scenario 16/16: K30_hiVar_int_imb ...

Each scenario is replicated $R = 1{,}500$ times. For coverage MCSE ≤ 0.6 pp at the nominal 95% level, $R \geq 1{,}584$; $R = 1{,}500$ gives MCSE ≈ 0.56 pp. Treatment is assigned via balanced allocation within each site (equal numbers of treatment and control participants, or within ± 1 when n_i is odd). All random number generation uses a fixed seed (`set.seed(20260310)`) and the `L'Ecuyer-CMRG` RNG for reproducibility.

3 Results

3.1 Summary of performance metrics

3.2 Bias

3.3 Power and coverage

3.4 Impact of imbalanced site sizes

4 Discussion

The simulation results can be organized around the four research questions motivating this study.

Modeling site effects. Consistent with[7] and[8], the random effects model attained nominal or near-nominal coverage and the highest power across nearly all scenarios. The fixed effects model performed comparably when the number of sites was small ($K = 10$) but lost efficiency with $K = 30$ due to the large number

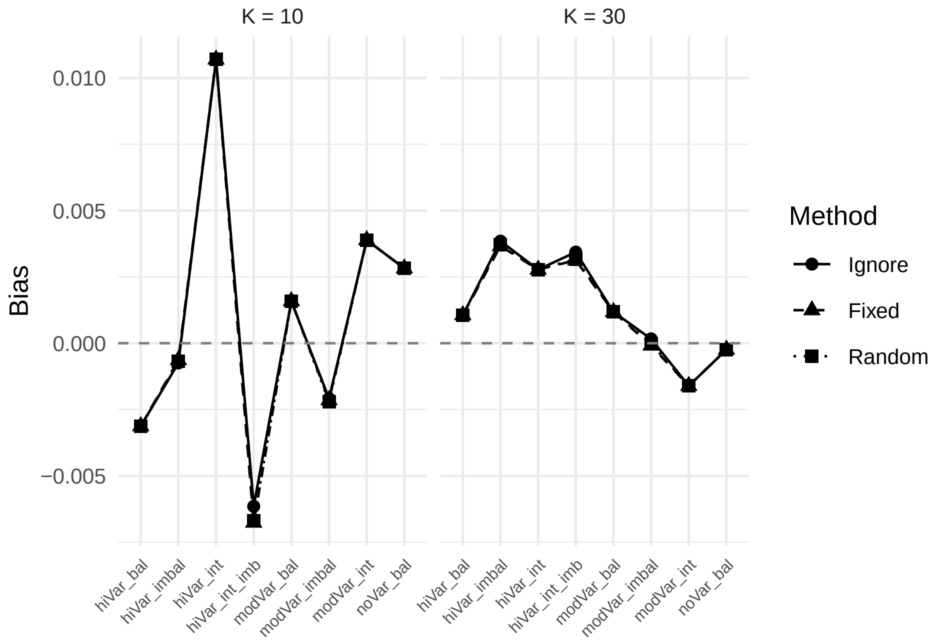

Figure 1: Bias of the treatment effect estimate across scenarios and analytic methods. Dashed line at zero indicates no bias.

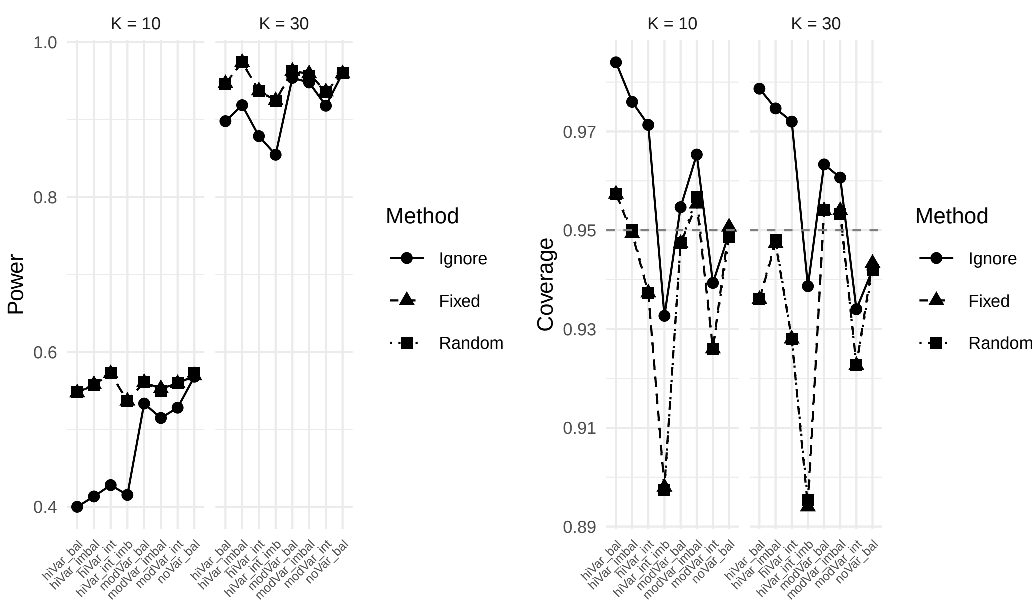

Figure 2: Power (left) and coverage probability (right) across scenarios. Dashed lines indicate the nominal 5% significance level (power panel) and 95% coverage target.

Table 1: Simulation results: bias, standard error, power, and coverage by scenario and analytic method. True treatment effect = 0.30; R = 1500 replications per scenario.

Scenario	Method	Bias	Emp. SE	Model SE	Power	Coverage	Conv.
K10 hiVar bal	Fixed	-0.003	0.140	0.141	0.547	0.957	100.0%
K10 hiVar bal	Ignore	-0.003	0.140	0.169	0.400	0.984	100.0%
K10 hiVar bal	Random	-0.003	0.140	0.141	0.548	0.957	100.0%
K10 hiVar imbal	Fixed	-0.001	0.141	0.141	0.558	0.949	100.0%
K10 hiVar imbal	Ignore	-0.001	0.142	0.167	0.413	0.976	100.0%
K10 hiVar imbal	Random	-0.001	0.141	0.141	0.557	0.950	100.0%
K10 hiVar int	Fixed	0.011	0.148	0.142	0.572	0.937	100.0%
K10 hiVar int	Ignore	0.011	0.148	0.171	0.428	0.971	100.0%
K10 hiVar int	Random	0.011	0.148	0.142	0.573	0.937	100.0%
K10 hiVar int imb	Fixed	-0.007	0.173	0.142	0.537	0.898	100.0%
K10 hiVar int imb	Ignore	-0.006	0.173	0.168	0.415	0.933	100.0%
K10 hiVar int imb	Random	-0.007	0.173	0.142	0.537	0.897	100.0%
K10 modVar bal	Fixed	0.002	0.141	0.141	0.561	0.947	100.0%
K10 modVar bal	Ignore	0.002	0.141	0.148	0.533	0.955	100.0%
K10 modVar bal	Random	0.002	0.141	0.141	0.561	0.947	100.0%
K10 modVar imbal	Fixed	-0.002	0.139	0.141	0.553	0.955	100.0%
K10 modVar imbal	Ignore	-0.002	0.139	0.147	0.515	0.965	100.0%
K10 modVar imbal	Random	-0.002	0.139	0.141	0.550	0.957	100.0%
K10 modVar int	Fixed	0.004	0.156	0.142	0.559	0.926	100.0%
K10 modVar int	Ignore	0.004	0.156	0.149	0.528	0.939	100.0%
K10 modVar int	Random	0.004	0.156	0.142	0.559	0.926	100.0%
K10 noVar bal	Fixed	0.003	0.140	0.142	0.569	0.951	100.0%
K10 noVar bal	Ignore	0.003	0.140	0.142	0.568	0.949	100.0%
K10 noVar bal	Random	0.003	0.140	0.141	0.573	0.949	100.0%
K30 hiVar bal	Fixed	0.001	0.082	0.082	0.947	0.936	100.0%
K30 hiVar bal	Ignore	0.001	0.082	0.099	0.898	0.979	100.0%
K30 hiVar bal	Random	0.001	0.082	0.082	0.947	0.936	100.0%
K30 hiVar imbal	Fixed	0.004	0.081	0.082	0.974	0.947	100.0%
K30 hiVar imbal	Ignore	0.004	0.081	0.098	0.919	0.975	100.0%
K30 hiVar imbal	Random	0.004	0.081	0.082	0.974	0.948	100.0%
K30 hiVar int	Fixed	0.003	0.092	0.082	0.937	0.928	100.0%
K30 hiVar int	Ignore	0.003	0.092	0.100	0.879	0.972	100.0%
K30 hiVar int	Random	0.003	0.092	0.082	0.937	0.928	100.0%
K30 hiVar int imb	Fixed	0.003	0.100	0.082	0.925	0.894	100.0%
K30 hiVar int imb	Ignore	0.003	0.101	0.099	0.855	0.939	100.0%
K30 hiVar int imb	Random	0.003	0.100	0.082	0.924	0.895	100.0%
K30 modVar bal	Fixed	0.001	0.079	0.082	0.963	0.954	100.0%
K30 modVar bal	Ignore	0.001	0.079	0.086	0.954	0.963	100.0%
K30 modVar bal	Random	0.001	0.079	0.082	0.963	0.954	100.0%
K30 modVar imbal	Fixed	0.000	0.081	0.082	0.959	0.954	100.0%
K30 modVar imbal	Ignore	0.000	0.081	0.085	0.948	0.961	100.0%
K30 modVar imbal	Random	0.000	0.081	0.082	0.956	0.953	100.0%
K30 modVar int	Fixed	-0.002	0.091	0.082	0.936	0.923	100.0%
K30 modVar int	Ignore	-0.002	0.091	0.086	0.918	0.934	100.0%
K30 modVar int	Random	-0.002	0.091	0.082	0.936	0.923	100.0%
K30 noVar bal	Fixed	0.000	0.082	0.082	0.959	0.943	100.0%
K30 noVar bal	Ignore	0.000	0.082	0.082	0.960	0.942	100.0%
K30 noVar bal	Random	0.000	0.082	0.081	0.960	0.942	100.0%

of nuisance parameters. Ignoring site effects produced unbiased point estimates but inflated standard errors when site-level variance was present, leading to conservative inference and reduced power. These findings support the random effects model as a robust default for multicenter RCTs, as recommended by[13] and[1].

Failure to adjust for stratification. The 'ignore site' method effectively represents a failure to adjust for the stratification variable (site) used in randomization. The overcoverage and power loss observed under this method confirm the findings

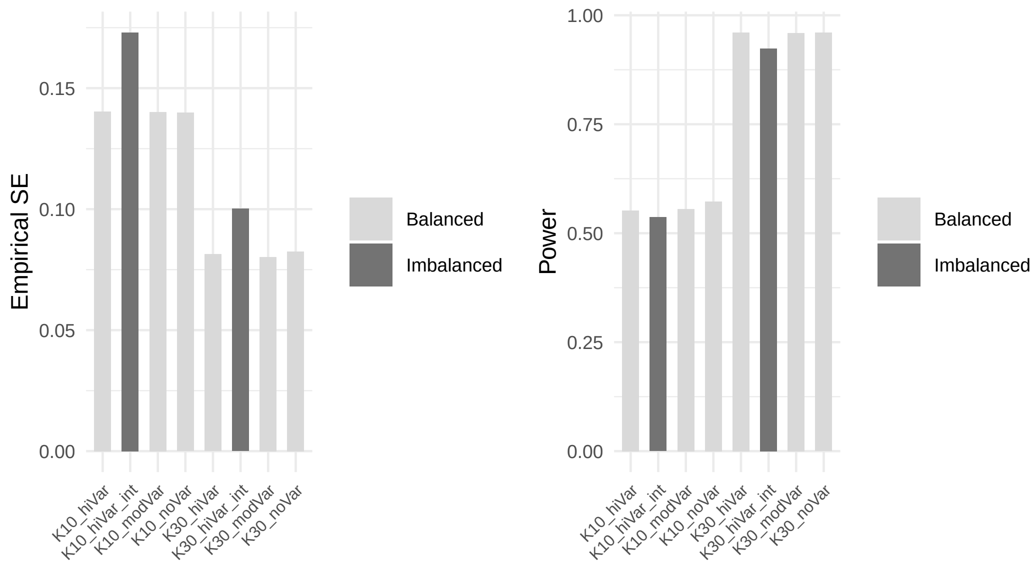

Figure 3: Comparison of balanced versus imbalanced site sizes for the random effects method. Panels show empirical standard error (left) and power (right).

of[10]: when randomization is stratified by center, the analysis must account for center to obtain valid inference. The magnitude of the power penalty depends on the ICC: with no site-level variance, ignoring site is harmless, but as σ^2_α increases, the penalty becomes substantial.

Treatment-by-site interaction. Introducing random treatment-by-site interaction ($\sigma^2_\gamma = 0.04$) increased the variability of treatment effect estimates across replications for all methods. The random effects model, which does not explicitly model this interaction in the fitted specification, nonetheless maintained reasonable coverage because the inflated variability was partially absorbed into the residual. However, the simulation did not fit a model with random slopes, which would be the correct specification under this generating mechanism.[4] and[12] have argued that some degree of treatment-by-center interaction is ubiquitous, and that the power to detect it is typically inadequate. Researchers should plan for its presence rather than test for its absence.

Imbalanced site sizes. The comparison of balanced versus imbalanced site sizes reveals that for the random effects model, moderate imbalance (CV ≈ 1.1) produces only a modest increase in empirical standard error and a correspondingly small decrease in power. This finding is consistent with[15] and[16], who showed that the efficiency loss from unequal site sizes is generally small unless the imbalance is extreme. The more concerning scenario is when site size is confounded with site characteristics – a possibility not simulated here but discussed by[12] and[14]. Under informative site sizes, where larger sites systematically differ from smaller sites, all methods may produce biased estimates of the population-average treatment effect.

Limitations. This simulation is restricted to continuous outcomes analyzed with normal-theory models. Binary and time-to-event outcomes raise additional complications, including separation in fixed effects logistic regression with many small sites[5,17,19]. The simulation does not incorporate dropout, non-compliance, or informative site sizes. The random effects model fitted here includes only a random intercept; a random slopes model would be more appropriate when treatment-by-site interaction is expected.

5 Future research

Several directions warrant further investigation:

1. **Random slopes models.** Extending the simulation to include a random treatment slope ($\delta + \gamma_i$ with γ_i estimated) would clarify when and how much the random slopes model outperforms the random intercept model under genuine treatment heterogeneity.

2. **Non-normal outcomes.** Binary and count outcomes require generalized linear mixed models or GEE, and the relative performance of methods may differ from the continuous case, particularly with sparse data per site[17,19].

3. **Informative site sizes.** If site size is associated with site-level treatment effect (e.g., expert centers enroll more patients and also deliver more effective treatment), standard methods may yield biased population-average estimates. Inverse-probability weighting by site or joint modeling of enrollment and outcome merit investigation[14].

4. **Small-sample corrections.** With few sites, the Wald-type inference from `lmer` may be anticonservative. Kenward-Roger or Satterthwaite denominator degrees of freedom corrections (available in `lmerTest`) should be evaluated across the same scenario grid.

5. **Bayesian approaches.** Hierarchical Bayesian models offer partial pooling of site-specific treatment effects, which may outperform frequentist random effects models when the number of sites is very small[4].

6. **Sample size planning.** Tools that incorporate the design effect from multicenter structure, including anticipated ICC and site size imbalance, into prospective sample size calculations would improve trial planning[16,20].

6 References

6.1 Morris et al. (2019) ADEMP Compliance

This simulation study was audited against the reporting standards proposed by Morris, White, and Crowther[18]. The full audit is at `docs/morris-audit-2026-04-17.md`.

Verdict: Partially compliant.

Key gaps identified:

- No Monte Carlo SE on any performance estimate in `summarize_simulation()`.
- `R = 1500` fixed without MCSE-based derivation.
- `filter(!is.na(est))` silently drops non-converged replications, breaking paired comparisons.

A remediation plan is documented in the audit file and will be executed in a subsequent revision.

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